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REVIEW

AI-driven hazard assessment: a comprehensive study of machine and deep learning approaches

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Abstract
Hazard assessment, whether for natural or man-made disasters, plays a critical role in minimizing the impact of environmental threats and protecting human lives and infrastructure. Traditional methods of hazard assessment often struggle to keep pace with the increasing complexity and frequency of hazard events. However, the integration of machine learning (ML) and deep learning (DL) techniques has revolutionized this field, offering powerful tools for data-driven predictions, risk analysis, and real-time decision-making. These AI-driven approaches enable the analysis of large-scale datasets, allowing for the identification of hidden patterns and correlations in hazard events, and improving the accuracy and scalability of hazard prediction models. In this study, we explore the applications of ML and DL in hazard assessment, highlighting their effectiveness in areas such as flood prediction, earthquake forecasting, and wildfire management. Machine learning techniques, including supervised and unsupervised learning, are used for predictive analysis, anomaly detection, and classification tasks, while deep learning models, particularly convolutional and recurrent neural networks, excel in handling spatial and temporal data such as satellite images and sensor data. Furthermore, hybrid models combining ML and DL techniques are discussed for their potential to enhance the robustness and accuracy of hazard predictions. The study also addresses the challenges of implementing AI in hazard assessment, including data quality issues, high computational costs, and the ethical implications of automated decision-making. Despite these challenges, the future of AI in hazard assessment looks promising, with innovations such as autonomous sensor networks, real-time data processing, and AI-powered global collaboration systems concrete the way for more effective hazard monitoring and disaster preparedness.

Keywords Deep learning · Hazard assessment · Machine learning · Predictive modeling · Real-Time monitoring · Risk analysis

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Introduction

Hazards, both natural and man-made, pose significant threats to human lives, infrastructure, and the environment. Predicting and assessing these hazards accurately has become a critical focus in scientific research, as traditional methods often struggle to keep pace with the increasing complexity of environmental and anthropogenic changes (Stalhandske et al., 2024). Over the past decade, advancements in artificial intelligence, particularly in machine learning (ML) and deep learning (DL), have transformed hazard assessment methodologies. These technologies provide powerful tools for analyzing large-scale data, identifying patterns, and making predictions that can mitigate the impact of disasters (Maki et al., 2024).

This section explores the fundamental principles of machine learning and deep learning and their applications in

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hazard assessment. By automating the extraction of meaningful features from complex datasets, these techniques allow for more accurate, timely, and scalable prediction models than conventional approaches. This introductory discussion lays the groundwork for understanding how ML and DL contribute to the evolving field of hazard assessment and how they have been successfully implemented in various domains.

Overview of hazards

Natural and anthropogenic hazards threaten human life, infrastructure, and the environment. Natural hazards, such as earthquakes, floods, hurricanes, and wildfires, are driven by environmental and geological forces, while anthropogenic (man-made) hazards include industrial accidents, pollution, and infrastructural failures. Each type of hazard brings unique challenges in prediction, mitigation, and response (Skoulidou & Kazantzi, 2025). The impact of these hazards can be catastrophic, often leading to loss of life, economic disruption, and environmental degradation. Understanding the underlying mechanisms and causes of these hazards is crucial for developing effective mitigation strategies (Figueiredo et al., 2024). Natural hazards are often triggered by complex interactions between environmental factors, including climate change, tectonic activity, and hydrological cycles. Anthropogenic hazards, on the other hand, are frequently associated with human error, negligence, or unintended consequences of industrial processes (Firdaus et al., 2024). Therefore, hazard assessment involves not only predicting the occurrence of these events but also understanding their potential consequences, which may vary based on geographic, socio-economic, and technological contexts.

Importance of hazards assessment

Hazards assessment plays a pivotal role in risk management and disaster preparedness. Accurate hazard assessment enables governments, organizations, and communities to allocate resources efficiently, design early warning systems, and implement disaster preparedness strategies to mitigate the risks posed by potential hazards (Nabil et al., 2024). The primary goal of hazard assessment is to minimize the impacts of disasters through preemptive measures that reduce vulnerability and enhance resilience. In the context of risk management, hazard assessment serves as a foundation for understanding the likelihood and severity of potential hazard events. This knowledge can inform land-use planning, emergency response protocols, infrastructure development, and insurance policies, among other applications. Hazards assessment also plays a critical role in the allocation of disaster relief resources and the prioritization of areas for intervention (Shao et al., 2024). For instance, accurate flood hazard maps can guide the construction of levees and flood barriers, while earthquake-prone zones may require stricter building codes.

Moreover, hazard assessment is integral to disaster preparedness and response. By predicting the timing and location of hazard events, authorities can issue early warnings, enabling timely evacuations and preventive actions. Early warning systems have proven invaluable in mitigating the loss of life during events such as tsunamis, hurricanes, and floods (Sun et al., 2024). The ability to forecast hazard events in advance significantly enhances the effectiveness of emergency response efforts, reducing the overall impact on communities. Figure 1 illustrates key elements of hazard assessment and highlights its importance in efficient resource allocation, early warning systems, preemptive disaster measures, infrastructure planning, and disaster relief prioritization, hazard event forecasting, and reducing overall disaster impact.

Traditional approaches to hazards assessment

As shown in Fig. 2, historically hazard assessment has relied on traditional methods, including statistical models, expert systems, and simulation techniques. These approaches, while effective in certain contexts, face limitations in handling the complex and dynamic nature of modern hazard events. Traditional methods often require extensive historical data and may not generalize well to novel or rapidly changing environmental conditions.

Statistical methods have been widely used for hazard assessment, particularly in the prediction of natural hazards like floods and earthquakes. For example, frequency analysis of historical data can be used to estimate the probability of future hazardous events (Lawal et al., 2024). However, these models often assume stationary conditions, which may not account for changing environmental factors such as climate change or urbanization. Additionally, the reliance on historical data may limit the ability to predict unprecedented or rare events (Yasmin et al., 2025a, 2025b). Furthermore, expert systems rely on the knowledge and experience of domain experts to develop rule-based models for hazard assessment. These systems are particularly useful in the absence of sufficient data, as they incorporate expert judgment and heuristic rules to predict hazard events. However, expert systems can be biased by the subjective opinions of the experts and may struggle to handle complex, nonlinear interactions between variables (Zhao et al., 2024). Moreover, simulation models commonly replicate hazard events and predict their impacts. For example, hydrological models simulate rainfall-runoff processes to predict flood extents,

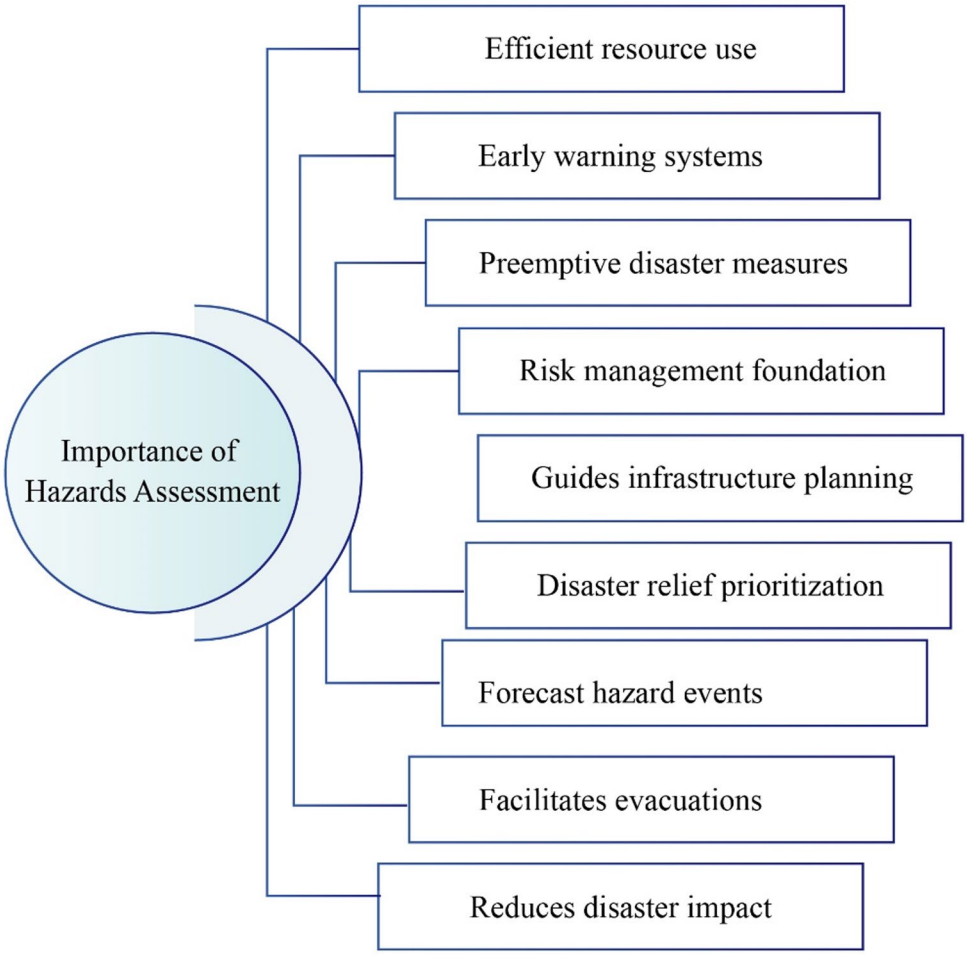


Fig. 1 Illustrates key elements of hazard assessment and highlights its importance in efficient resource allocation, early warning systems, preemptive disaster measures, infrastructure planning, disaster relief prioritization, hazard event forecasting, and reducing overall disaster impact

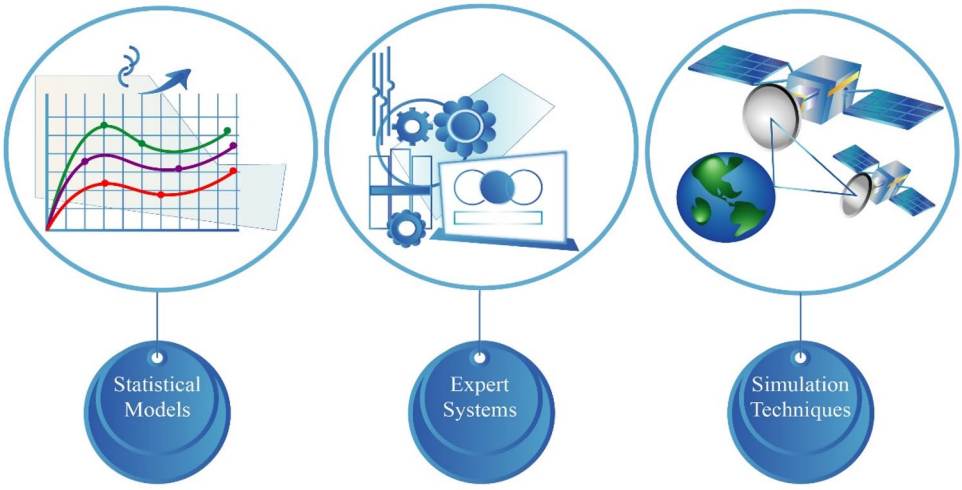


Fig. 2 Illustrates traditional approaches to hazard assessment including statistical models, expert systems, and simulation techniques

while seismic models estimate ground shaking intensity during earthquakes (Jaberzadeh et al., 2024). Although simulation techniques provide valuable insights into the mechanics of hazardous events, they require accurate and detailed input data. Errors in data collection or model assumptions can lead to significant discrepancies between simulated outcomes and real-world events.

Despite their usefulness, traditional methods' limitations have become increasingly evident as the complexity of hazardous events grows. The rise of large-scale, multidimensional data from sensors, satellite imagery, and other

sources has highlighted the need for more sophisticated approaches capable of integrating diverse datasets and adapting to dynamic environmental conditions. Traditional methods often struggle to process such vast and complex datasets, and their predictive power may degrade in novel or evolving scenarios.

Limitations of traditional methods

One of the primary limitations of traditional hazard assessment methods is their reliance on historical data, which may not adequately represent future conditions, particularly in the context of climate change and rapid urbanization. Many statistical models assume that the frequency and intensity of hazardous events will remain consistent over time, an assumption that may not hold as environmental and socio-economic conditions evolve (Zeng et al., 2024). Additionally, traditional methods often operate in silos, focusing on specific types of hazards without considering their interdependencies. For instance, an earthquake can trigger secondary hazards such as tsunamis or landslides, which may not be adequately captured by models that focus solely on seismic activity (Shi et al., 2024).

Furthermore, traditional models typically require a high degree of human intervention, either in the form of expert knowledge or manual data processing. This reliance on human expertise introduces potential biases and limits scalability, especially in regions where access to experts or high-quality data is limited (Anderson et al., 2024). Moreover, traditional methods often struggle to provide real-time predictions, as they require extensive computation or manual adjustment of parameters, making them less effective for early warning systems and rapid disaster response. Given these limitations, there is a growing recognition of the need to adopt more advanced methods for hazard assessment. Machine learning and deep learning techniques (Hasan et al. 2025b, 2026), in particular, offer promising solutions by enabling the automated analysis of large and complex datasets, improving predictive accuracy, and providing real-time insights (Mandal & Mandal, 2024). These approaches can complement and enhance traditional methods, leading to more robust and adaptive hazard assessment frameworks.

Machine learning and deep learning for hazards prediction

In this section, we explore the pivotal role of ML and DL in enhancing hazard prediction methodologies. We delve into the various algorithms and architectures employed in hazard assessment, examine their applications in predicting specific hazards such as earthquakes and floods, and highlight the emerging hybrid approaches that combine the strengths of both ML and DL. Furthermore, we address the challenges associated with implementing these advanced techniques in real-world scenarios. The increasing complexity and unpredictability of hazard events, coupled with the growing availability of large-scale environmental data, have prompted a shift from traditional hazard assessment methods to more sophisticated approaches like machine learning and deep learning. These techniques have demonstrated substantial promise in their ability to identify patterns, predict future hazard events, and optimize early warning systems. By leveraging the power of algorithms to process vast datasets, machine learning and deep learning have enhanced the accuracy, speed, and adaptability of hazard prediction systems. The integration of ANNs in civil engineering applications has been extensively demonstrated, providing valuable insights into structural optimization and predictive analytics (Kaveh, 2024).

Overview of machine learning and deep learning algorithms

Machine learning refers to a set of techniques that enable computers to learn from data and make predictions without being explicitly programmed for specific tasks. In the context of hazard prediction, machine learning models can be trained to identify patterns in historical data and use them to predict future events.

ML and DL techniques offer a wide range of approaches for hazard prediction, each suited to different data types and prediction tasks (Saeed et al., 2024). For instance, decision trees and random forests provide both interpretability and accuracy, with random forests particularly excelling in handling complex datasets by combining multiple trees, which enhances prediction accuracy, especially for hazards like floods and wildfires. Similarly, support vector machines (SVMS) are highly effective for classifying seismic data, while k-nearest neighbors (KNN) leverage the proximity to past events, such as landslides, for prediction (Hasan et al., 2023). Moreover, gradient boosting machines (GBMS) improve model performance by correcting previous errors, making them valuable in wildfire risk assessment. On the deep learning side, convolutional neural networks (CNNs) are well-suited for processing spatial data, such as satellite images for flood detection. In contrast, recurrent neural networks (RNNs) and long short-term memory (LSTM) networks excel in handling sequential data like seismic readings or weather patterns (Khan et al., 2024). Additionally, auto encoders are useful for anomaly detection by reducing dimensionality and aiding in the identification of potential hazards. Lastly, generative adversarial networks (GANs) contribute by creating synthetic data to augment

small or imbalanced datasets, which is particularly beneficial for rare events like earthquakes or flash floods. Altogether, these techniques complement one another, offering versatile solutions to enhance the accuracy and reliability of hazard prediction models (Ma et al., 2024). Table 1 integrates both ML and DL techniques, providing a clear comparison of their applications in hazard prediction, strengths in handling various data types, and key limitations. It offers a comprehensive overview for readers looking to understand the advantages and challenges of these methods in different hazard contexts.

These algorithms, although capable of handling structured hazard data, often require manual feature engineering and may not capture the intricate spatiotemporal relationships inherent in many natural and man-made hazards. As hazard datasets grow in size and complexity, deep learning models provide a powerful alternative by automating feature extraction and delivering state-of-the-art predictive performance.

Table 1 Summary of machine learning and deep learning techniques for hazard prediction

Technique	Application	Strengths	Limitations
Decision trees	Flood prediction, landslides	Easy to interpret, handles complex data	Prone to overfitting, especially on small datasets
Random forests	Earthquake prediction, flood risk	High accuracy, reduces overfitting	Computationally expensive for large datasets
Support vector machines (SVM)	Seismic activity prediction	Effective in high-dimensional spaces	Requires careful tuning, sensitive to noise
k-nearest neighbors (k-NN)	Landslide prediction	Simple to implement, non-parametric	Struggles with large datasets, computationally expensive
Gradient boosting machines (GBMs)	Wildfire risk prediction	High accuracy, minimizes prediction errors	Can overfit, requires extensive computation
Convolutional neural networks (CNNs)	Flood detection from satellite imagery	Excellent for image-based data	Requires large datasets and computation
Recurrent neural networks (RNNs)	Seismic activity prediction, weather patterns	Good for sequential data, time series	Difficult to train, prone to vanishing gradients
Long short-term memory (LSTM)	Flood prediction, earthquake analysis	Good for long-term dependencies	Computationally intensive, slower to train
Autoencoders	Anomaly detection in hazard datasets	Effective in dimensionality reduction	Needs careful tuning, risk of overfitting
Generative adversarial networks (GANs)	Synthetic data generation for rare hazards	Good for imbalanced datasets	Complex to train, prone to mode collapse

Deep learning architectures for hazards prediction

Deep learning, a subset of machine learning, focuses on models with multiple layers of neurons, commonly referred to as neural networks. These models have demonstrated remarkable success in handling unstructured data, such as images, video, and time-series data, making them ideal for prediction where complex patterns across space and time are prevalent. Two primary deep learning architectures commonly applied in hazard prediction are CNNs and RNNs. CNNs are highly effective in analyzing spatial data, such as satellite images, aerial photos, and geographic maps. The architecture of CNNs allows them to automatically learn spatial hierarchies of features, making them suitable for detecting patterns in images related to hazards such as wildfires, floods, and land-use changes. For example, by analyzing vegetation indices and topographical features, CNNs can be trained on satellite imagery to predict flood extents after heavy rainfall or detect regions prone to wildfire outbreaks (Wang et al., 2024). RNNs, and their variant LSTM networks, are specialized for processing sequential data. In hazard prediction, they are used to model time-series data such as rainfall patterns, temperature changes, or seismic readings. RNNs excel in capturing temporal dependencies and trends, making them valuable for forecasting hazards that evolve, such as hurricanes or droughts. LSTMs are particularly effective in retaining long-term dependencies, allowing for the prediction of hazard events with long-duration precursors, like heatwaves or droughts (Yang et al., 2024). The flexibility of deep learning in handling both spatial and temporal data makes it a vital tool in modern hazard prediction systems. These models can be trained on vast datasets from diverse sources to provide more accurate and timely predictions than traditional models. Figure 3 illustrates the scenario of machine learning and deep learning applications on earthquake and flood prediction.

ML and DL have been increasingly applied to earthquake and flood prediction due to their capacity to analyze multi-modal data and identify subtle patterns that are often undetectable by traditional methods. Predicting earthquakes involves analyzing seismic signals, geological data, and historical records to detect patterns that may indicate an impending earthquake. Machine learning models such as decision trees and random forests have been used to classify seismic regions based on their likelihood of experiencing earthquakes. However, the incorporation of deep learning techniques, particularly CNNs, has enabled researchers to

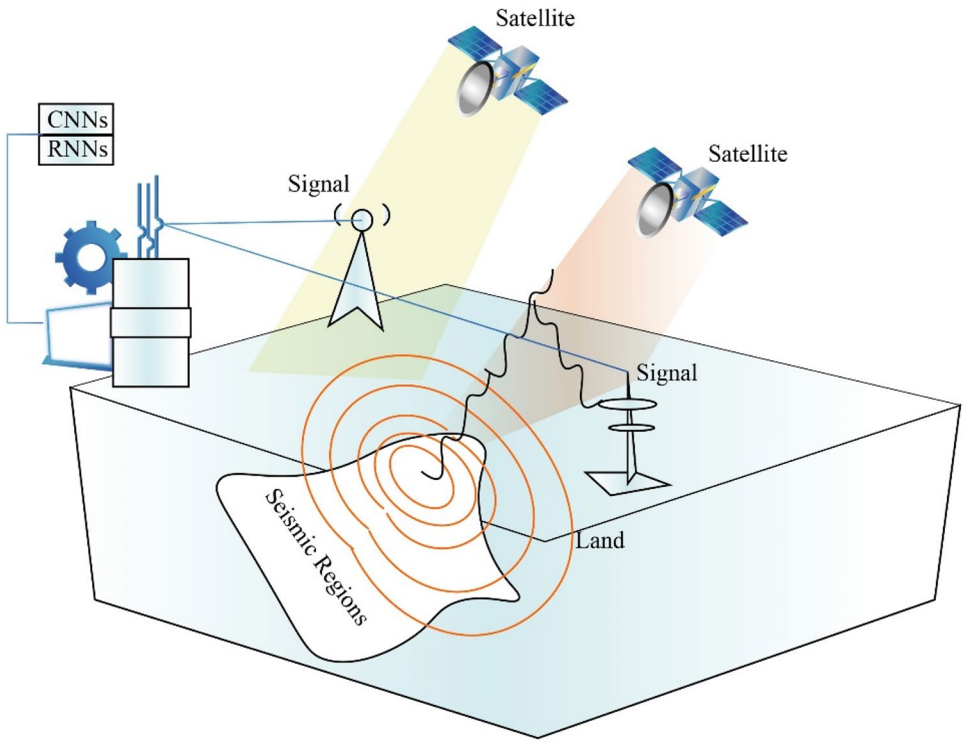


Fig. 3 Illustrates the scenario of ML and DL applications on earthquake and flood prediction. Where earthquakes involve analyzing seismic signals, geological data, and historical records and flood prediction requires the integration of various data sources, such as meteorological data, river discharge records, and topographical maps

analyze spatial features in seismic maps and automatically extract predictive features (Joshi et al., 2024). RNNs, on the other hand, have been used to model time-series data from seismic sensors, helping to predict earthquake occurrence based on real-time monitoring. On the other hand, flood prediction requires the integration of various data sources, such as meteorological data, river discharge records, and topographical maps. Machine learning models have been applied to predict flood risks by analyzing historical flood events and precipitation patterns. Random forests, for example, have been used to classify regions based on flood vulnerability (Dotse et al., 2024). In recent years, CNNs have been applied to satellite imagery to identify flood-prone areas and monitor flood extents during extreme weather events. Additionally, RNNs have been employed to forecast river discharge levels and rainfall intensity, providing more accurate flood predictions with lead times that are critical for early warning systems.

Challenges in applying machine learning and deep learning to hazards prediction

Hazards prediction models rely heavily on large, high-quality datasets. However, data on past hazard events may be scarce, incomplete, or unreliable, particularly in developing regions. Additionally, environmental data is often noisy or subject to measurement errors, which can negatively impact model performance (Krichen et al., 2024). While DL models, particularly CNNs and RNNs, have shown excellent predictive performance, their complexity often makes them difficult to interpret. This lack of transparency can pose challenges when these models are used in decision-making processes for disaster preparedness and response (Albahri et al., 2024). Moreover, DL models require significant computational resources for training and prediction, especially when dealing with large-scale, high-dimensional datasets such as satellite imagery or sensor networks. This complexity can be a barrier to deploying these models in real-time hazard assessment systems (Pwavodi et al., 2024). Figure 4 illustrates the challenges in applying machine learning and deep learning to hazards prediction.

ML and DL have revolutionized the field of hazard prediction by offering sophisticated models that can process vast and complex datasets with high precision. These methods have demonstrated their ability to predict various hazard events, such as earthquakes and floods, by identifying patterns within both structured and unstructured data. With innovations in hybrid modeling and the integration of spatial and temporal data, the potential for improving hazard assessments is immense. However, challenges such as data quality, model interpretability, and computational demands remain, requiring continued research and development to fully realize the potential of these technologies in disaster risk reduction and early warning systems.

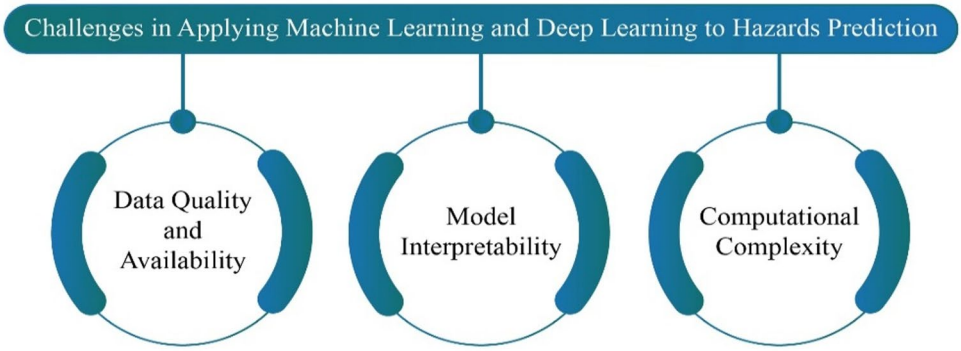


Fig. 4 Illustrates the challenges in applying machine learning and deep learning to hazards prediction

Date	Temperature (°C)	Precipitation (mm)	Humidity (%)	Wind Speed (km/h)
2024-01-01	12.5	5.6	78	15
2024-01-02	NaN	0.0	80	12
2024-01-03	13.1	NaN	75	20
2024-01-04	11.7	2.3	NaN	18
2024-01-05	14.0	0.0	77	NaN
2024-01-06	NaN	NaN	79	16
2024-01-07	12.9	3.5	76	NaN

Applied KNN, it considers the most similar rows to fill in missing values, making the imputed values more contextually relevant.

Date	Temperature (°C)	Precipitation (mm)	Humidity (%)	Wind Speed (km/h)
2024-01-01	12.5	5.6	78	15
2024-01-02	12.3	0.0	80	12
2024-01-03	13.1	0.1	75	20
2024-01-04	11.7	2.3	75.1	18
2024-01-05	14.0	0.0	77	18.1
2024-01-06	14.1	0.1	79	16
2024-01-07	12.9	3.5	76	16.1

Fig. 5 Illustrates the scenario of the hazard dataset's missing data handling with KNN. Where the NaN value is replaced by its nearest neighbor value

Data preprocessing for hazards prediction

In hazard prediction, the quality and relevance of data play a crucial role in the performance of ML and DL models. Data preprocessing, a critical step before model training, involves cleaning, transforming, and organizing raw data to ensure that models receive meaningful and structured input. Proper preprocessing not only enhances model performance but also addresses common challenges such as noise, missing values, and imbalanced datasets, which are often present in hazard data. This section delves into the key techniques and methodologies for effective data preprocessing in the context of hazard prediction.

Data cleaning and handling missing data

Raw data collected from diverse sources, such as satellite sensors, IoT devices, or historical records, often contains errors, inconsistencies, or incomplete entries. Data cleaning is the process of detecting and correcting such inaccuracies (Hasan et al., 2025a, 2025b). In hazard prediction, ensuring the integrity of data is paramount, as even minor errors can lead to significant mispredictions. Hazard datasets may contain duplicate entries, especially in time-series data from sensors or crowd-sourced reports. Duplicates can skew the training process, leading to overfitting or bias. Moreover, missing data is common in hazardous datasets due to sensor malfunctions, incomplete records, or transmission errors. Approaches to handle missing data include imputation methods such as mean substitution, k-nearest neighbors (KNN) imputation, or more advanced approaches like data augmentation or generative models (Ishibashi, 2024). Alternatively, records with missing values can be excluded, though this may reduce the overall dataset size. Figure 5 illustrates the scenario of the hazard dataset's missing data handling with KNN.

Therefore, proper data cleaning is essential for ensuring the accuracy and reliability of hazard prediction models.

By removing duplicates and addressing missing values, the dataset becomes more consistent and robust, leading to better performance in predicting hazards.

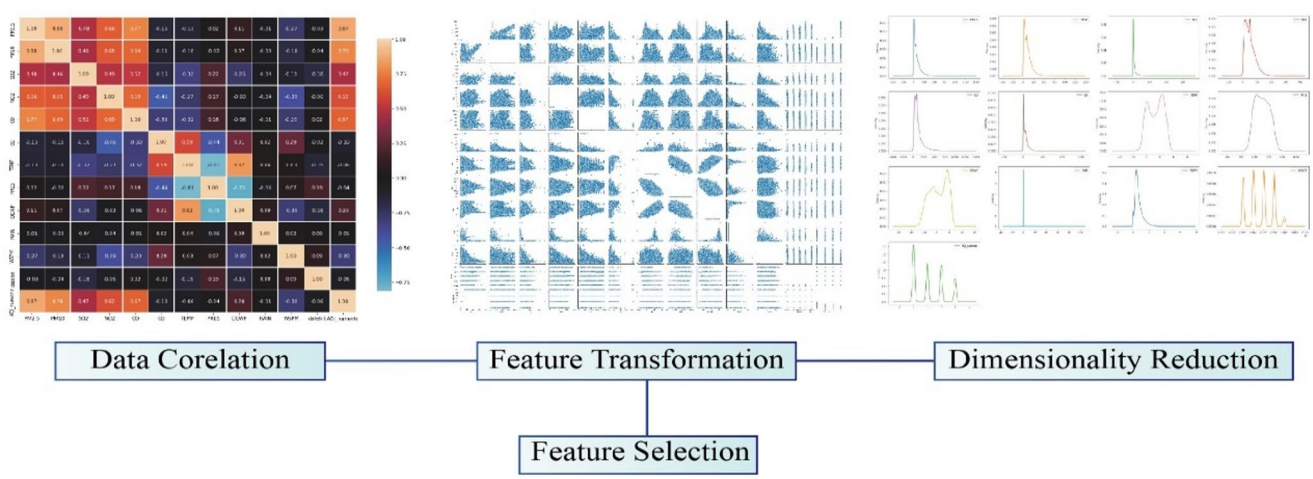
Feature engineering and selection

Feature engineering involves transforming raw data into meaningful features that help the model capture important patterns. For hazard prediction, features could include geographical coordinates, weather conditions, past hazard occurrences, or environmental variables such as soil moisture or vegetation indices. The quality of these features significantly affects the model's predictive capabilities. Feature Transformation involves scaling and normalizing data to ensure that different features contribute equally to model training. For example, sensor data from rainfall and temperature sensors may need to be scaled into the same range to prevent any single feature from dominating the learning process (Chai et al., 2024). In cases where datasets have a large number of features, techniques such as principal component analysis (PCA) or t-SNE are applied to reduce dimensionality while retaining the most relevant information. This is crucial in hazard data, where multiple variables may be highly correlated, such as rainfall and humidity levels, which may add redundancy to the model. Additionally, selecting the most relevant features is essential for reducing model complexity and improving efficiency (Yasmin et al., 2023). Algorithms such as recursive feature elimination (RFE), mutual information, or regularization methods (help identify key predictive variables. For example, when predicting wildfires, features such as wind speed, humidity, and temperature may be more relevant than soil composition, helping models focus on the most significant factors. Figure 6 illustrates the scenario of the hazard dataset's feature engineering including data correlation, feature transformation, dimensionality reduction, and feature selection.

Effective feature engineering is essential for improving hazard prediction models. By transforming raw data into relevant features, reducing dimensionality, and selecting the most pertinent variables, the models can focus on critical patterns that enhance predictive accuracy and overall performance. These techniques play a vital role in ensuring that the models can accurately assess risks and make informed predictions in various hazard scenarios.

Data normalization and standardization

Normalization and standardization are key steps in preparing data for ML and DL models, ensuring that input features contribute equally to the learning process. In hazard prediction, where variables such as temperature, rainfall, or elevation can have vastly different ranges, normalization



	PM2.5	PM10	SO2	NO2	CO	O3
Count	418,946	418,946	418,946	418,946	418,946	418,946
Mean	79.19028	104.1894	15.65343	50.59347	1212.644	57.11511
STD	80.0203	91.08931	21.47491	34.60304	1132.051	55.84823
Min	2	2	0.2856	1.0265	100	0.2142
25%	21	36	3	24	500	12
50%	55	82	7	45	900	45
75%	109	144	19	70	1500	80
Max	999	999	500	290	10,000	1071

Fig. 6 Illustrates the scenario of the hazard dataset's feature engineering including data correlation, feature transformation, dimensionality reduction, and feature selection

techniques are used to scale these values into a comparable range. Standardization, on the other hand, centers the data by subtracting the mean and dividing by the standard deviation, making it easier for models to learn patterns without being skewed by large or small values.

Min–max normalization rescales features to a range of [0, 1] or [− 1, 1]. This is particularly useful when working with sensor data, where different sensors might have varying ranges. Z-score standardizes features by transforming them to have a mean of 0 and a standard deviation of 1. This method is commonly applied to hazard datasets where the distribution of values varies significantly across different regions or sensors (Kim et al., 2024). The mean normalization formula and z-score normalization for a feature x' is shown in Eqs. (1, 2) as follows.

$$x' = \frac{x - mean(x)}{\max(\mathrm{x}) - \min(\mathrm{x})} \tag{1}$$

$$x' = \frac{x - mean(x)}{\mathrm{std}(\mathrm{x})} \tag{2}$$

Where x is the original feature value, $mean(x)$ is the mean of the feature x, $max(x)$ is the maximum value of the feature x, $min(x)$ is the minimum value of the feature x and $std(x)$ is the standard deviation of the feature x.

Normalization and standardization are critical processes in preparing hazard prediction data, ensuring that diverse input features contribute equally to model training. By employing techniques like min-max normalization and z-score standardization, data is effectively scaled and centered, allowing models to learn patterns more accurately. This preprocessing step enhances the models' ability to manage variations across different variables, ultimately leading to improved performance in predicting hazardous events.

Evaluation metrics and techniques for hazards prediction

Evaluating ML and DL models for hazard prediction is crucial for ensuring reliable and actionable outcomes. Selecting the right evaluation metrics becomes particularly important when dealing with imbalanced datasets, as hazard prediction often involves high-stakes decisions. Effective evaluation not only measures a model's predictive accuracy but also addresses the balance between false alarms and missed hazards, both of which are critical in real-world applications. This section delves into the key evaluation metrics and techniques commonly applied in hazard prediction, offering insights into their relevance and utility.

Accuracy, precision, and recall are essential metrics for evaluating classification models in hazard prediction. While accuracy indicates the proportion of correct predictions, it can be misleading in imbalanced datasets, which are common in this field. Precision assesses the correctness of positive predictions, whereas recall measures the ability to detect all actual hazard events. Wang et al. (2024a, 2024b) proposes an ML-based approach to analyze reservoir bank landslide reliability using MLP, CNN, and LSTM models, finding that CNN outperforms the others and that optimizing data quantity and ratio enhances prediction accuracy. Liu et al. (2024a, 2024b) performed the verhulst, GM (1,1), and saito models in predicting landslide life expectancy are analyzed using displacement data from four cases, with an ML-based ensemble system proposed to enhance accuracy, and a discredit index introduced to assess prediction quality, where higher values indicate lower quality. Striking a balance between these metrics is vital to avoid trade-offs, particularly in scenarios like flood prediction, where high recall is necessary to capture most events, and high precision helps reduce false alarms. Nguyen et al. (2024) to achieve this balance, the F1 score, the harmonic mean of precision, and recall have been employed. The F2 score prioritizes recall, which is significant in contexts like earthquake prediction, where missing hazards are more detrimental than false alarms.

Additionally, the ROC curve and AUC are valuable for evaluating model performance across different thresholds, helping to balance sensitivity and specificity. The ROC curve illustrates trade-offs between true and false positives, while a higher AUC indicates strong model discriminatory power. He et al. (2024) evaluated four ML algorithms XGBoost, RF, BP, and LSTM to create Geohazard Susceptibility Maps for the Greater Bay Area, finding that XGBoost outperformed the others (AUC = 0.89), while also investigating geohazard triggers and interactions using a vine copula model. Furthermore, a confusion matrix provides detailed insights into model predictions, distinguishing between true positives, true negatives, false positives, and false negatives. Özen and Souri (2024) develop an optimized ensemble bagged tree (OEBT) algorithm with ANOVA-based feature selection to rapidly process disaster data, achieving 97.9% accuracy, with confusion matrix analysis showing the precise classification of major disasters (100% true positive rate) but lower accuracy for fire management (36.4%). Moreover, cross-validation divides the dataset into subsets for more reliable performance assessment, particularly in temporally or spatially varied hazard data. And, bootstrapping, which samples with replacement, generates multiple training datasets, offering robust performance estimates, especially for rare hazards. Thanh et al. (2024) present groundwater quality maps using RF with cross-validation (RF-CV), RF with

Table 2 Evaluation metrics and techniques for hazard prediction models

Metric/technique	Description	Importance in hazard prediction
Accuracy	Proportion of correctly predicted instances	Can be misleading in imbalanced datasets; and may misrepresent model performance by showing high accuracy while failing to detect hazards
Precision	Proportion of true positive hazard predictions among all positive predictions	Important for minimizing false alarms, crucial in scenarios like flood prediction where unnecessary alerts can cause panic
Recall	Ability to detect all actual hazard events	Essential for ensuring most hazards are detected, particularly in high-risk scenarios like flood prediction
F1 score	Harmonic means of precision and recall	Provides a balanced assessment in scenarios where both false negatives and positives are critical, like flood prediction
F2 score	Similar to F1, but gives more weight to recall	Valuable in situations where detecting every possible hazard (e.g., earthquake prediction) is more critical than avoiding false alarms
ROC curve	Visual representation of the trade-off between true positives and false positives at various thresholds	Helps fine-tune model performance by allowing adjustments based on operational needs, particularly in wildfire prediction
AUC (area under the curve)	Summarizes a model's ability to distinguish between hazard and non-hazard events	A higher AUC indicates strong discriminatory power, which is essential for reliable hazard warnings
Confusion matrix	Breakdown of model predictions into true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN)	Provides insights into model performance, helping to identify types of errors and their potential impacts in high-stakes situations
Cross-validation	Technique that splits the dataset into multiple subsets for testing	Enhances model generalizability, especially useful when hazard data varies temporally or spatially
Bootstrapping	Sampling with replacement to generate multiple training datasets	Provides robust performance estimates, especially valuable for small or imbalanced datasets common in rare hazards

bootstrapping (RF-B), ANN with cross-validation (ANN-CV), and ANN with bootstrapping (ANN-B) models, showing that RF-CV and RF-B estimated around 94–95% of the area with good or very good groundwater quality, while ANN-CV and ANN-B predicted 93–94%, with minor variations in the moderate and poor quality zones across models. Table 2 provides a structured overview of key evaluation metrics and techniques used in hazard prediction models, emphasizing their relevance and implications in ensuring effective and reliable model performance.

Overall, applying these evaluation metrics and techniques allows researchers to critically assess the performance of machine learning and deep learning models in hazard prediction. These tools ensure that models are both accurate and reliable in high-risk scenarios, where even small errors can have significant consequences. Proper evaluation is a cornerstone in advancing predictive capabilities, leading to more effective hazard mitigation and preparedness systems.

Real-world applications of machine learning and deep learning in hazard prediction

The integration of ML and DL in hazard prediction has transformed the field, enabling more accurate, timely, and efficient forecasting of various hazard events. By harnessing large datasets and complex algorithms, these technologies offer innovative solutions across different domains, ranging from meteorology to public safety. This section explores several real-world applications of ML and DL in hazard prediction, highlighting their effectiveness and impact.

Natural disasters

Machine learning and deep learning techniques have shown remarkable efficacy in predicting various natural disasters, including floods, heatwaves, earthquakes, and wildfires. ML models, including regression techniques and random forests, have been utilized to predict flood events by analyzing historical rainfall data and river levels.

Dtissibe et al. (2024) compare machine learning and deep learning models 1D CNN, LSTM, and MLP for short- and long-term flood forecasting in Cameroon using temperature and rainfall data, finding that LSTM performs best for both short- and long-term predictions based on multiple performance metrics. Lee et al. (2024) employed an ML model and explainable AI to identify key teleconnection drivers for heatwaves in South Korea, finding that snow depth variability in the Gobi desert and Tianshan mountains are the most significant predictors, closely correlated with heatwave-conducive summer climate conditions. PreD-Net, a deep learning method for identifying precursory signals

of induced earthquakes, demonstrating strong generalization across different seismicity areas and showing potential for real-time warning strategies to mitigate risks during field operations proposed by Convertito et al. (2024). The increasing frequency and intensity of wildfires have necessitated the use of ML and DL for effective detection and management. Remote sensing through satellite imagery and drones equipped with ML algorithms can monitor vegetation health, weather conditions, and fire behavior, helping predict potential wildfire outbreaks. Real-time monitoring using deep learning models, particularly CNNs, analyzes data from various sensors to detect heat signatures indicative of wildfires, allowing for quicker response times and better resource allocation during fire events. Jonnalagadda and Hashim (2024) proposed a SegNet-based approach to enhance processing speed and accuracy in real-time wildfire detection from UAV/drone imagery, addressing the challenge of limited datasets by reducing feature maps for improved time resolution and detection capabilities. Figure 7 illustrates the scenario of predicting various natural disasters, including floods, heatwaves, earthquakes, and wildfires.

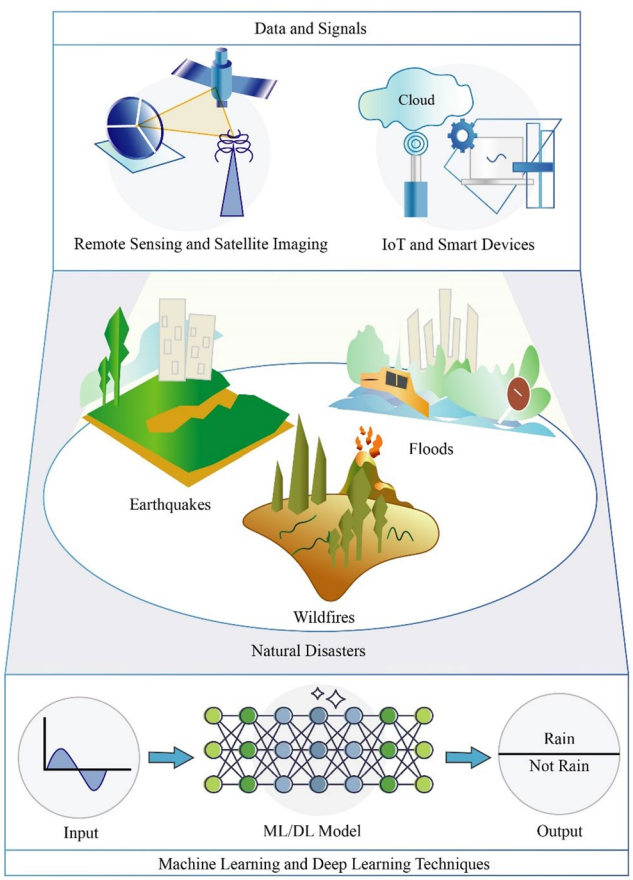


Fig. 7 illustrates the scenario of the predicting various natural disasters, including floods, heatwaves, earthquakes, and wildfires

The application of machine learning and deep learning in weather prediction and natural disasters has revolutionized our ability to forecast and respond to hazardous events. By utilizing advanced algorithms to analyze complex datasets, these technologies enhance the accuracy and timeliness of predictions for phenomena such as floods, tornadoes, and hurricanes. The integration of these predictive models not only improves disaster preparedness but also significantly reduces risks to human life and property. As the frequency of extreme weather events continues to rise, the role of ML and DL in mitigating their impacts will become increasingly critical.

Urban planning and infrastructure resilience

The application of ML and DL in hazard prediction extends to urban planning and infrastructure resilience, where data-driven insights can enhance community safety. These technologies enable cities to better anticipate, prepare for, and respond to various hazards, ensuring the sustainability of urban environments.

DL algorithms can analyze geographical, demographic, and environmental data to assess the risk levels of different areas to various hazards, such as flooding, earthquakes, and heatwaves. This risk assessment is crucial for informed decision-making regarding land use, zoning regulations, and infrastructure development. Alshayeb et al. (2024) developed DL models combined with Bayesian optimization and explainable AI to improve storm surge susceptibility prediction in Sagar Island, addressing the limitations of traditional methods and enhancing coastal management and disaster risk reduction. By analyzing factors such as population density, mobility patterns, environmental conditions, and historical outbreak data, these models can forecast potential outbreaks and identify high-risk areas (Hasan et al., 2025a, 2025b). This predictive capability is invaluable for public health strategies and resource allocation during health crises. Furthermore, implementing ML in smart city initiatives allows for real-time monitoring of infrastructure. Shuhan et al. (2024) advanced smart city cybersecurity through three initiatives: proposing a CycleGAN-based data architecture to counter false data injection attacks (FDIA), enhancing data integrity, and introducing an IoT concept to improve real-time monitoring and context-aware threat detection in intrusion detection systems (IDS).

Moreover, DL techniques can analyze urban heat patterns and identify areas most susceptible to extreme heat events. By optimizing urban designs and implementing green infrastructure solutions, cities can mitigate the effects of urban heat islands, improving public health outcomes during heat waves. Johannsen et al. (2024) applied DL models, specifically CNNs, to downscale land surface temperature and

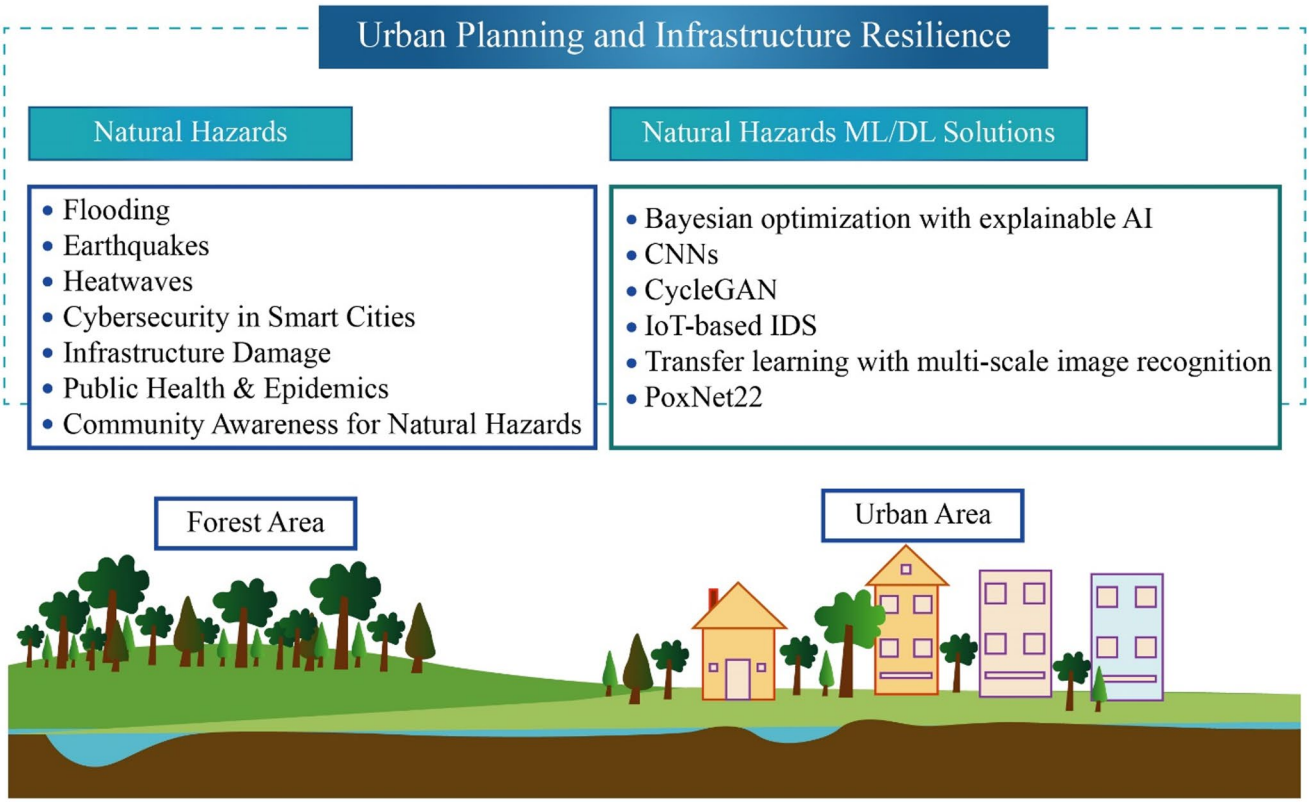


Fig. 8 Illustrates the scenario of the application of ML and DL in hazard prediction to urban planning and infrastructure resilience

predict maximum and minimum 2-m temperatures over Paris (2004–2022), comparing results with ERA5 data, and assessing the models' effectiveness in representing the urban heat island effect. Bai et al. (2024) proposed a deep learning-based multi-scale image damage recognition and assessment method for reinforced concrete structures, incorporating a damage image dataset, transfer learning, and a multi-scale assessment procedure, with an engineering case demonstrating its real-world feasibility. Similarly, optimizing the compressive strength of concrete using artificial neural networks combined with metaheuristic algorithms can significantly enhance the structural resilience of urban infrastructure, ensuring better resistance against hazards (Hosseini et al., 2023). In addition, the efficient training of artificial neural networks with metaheuristic algorithms has been utilized to accurately predict the strength of fiber-reinforced polymers (FRPs), which are critical for retrofitting and strengthening structures in hazard-prone urban environments (Kaveh & Khavaninzadeh, 2023). Furthermore, ML can facilitate community engagement in urban planning by analyzing natural hazard awareness and disaster risk reduction. This participatory approach ensures that urban development aligns with community needs, enhancing the effectiveness of hazard mitigation strategies. Bubeck et al. (2024) demonstrated the use of participatory theatre as a tool for natural hazard awareness and disaster risk reduction, in line with the Sendai Framework's call for community-based efforts, though quantitative evaluations remain limited. Additionally, DL can be used to develop personalized health interventions based on individual risk factors and health behaviors. By analyzing patient data, ML algorithms can identify individuals at high risk for certain diseases, enabling targeted preventive measures and early interventions. Yasmin et al. (2023) developed an image-processing model for diagnosing monkeypox, using transfer learning across six DL models, ultimately proposing "PoxNet22" after fine-tuning the best-performing model.

Figure 8 illustrates the scenario of the application of ML and DL in hazard prediction to urban planning and infrastructure resilience. The integration of ML and DL in urban planning and infrastructure resilience is transforming how cities prepare for and respond to hazards. By enabling data-driven risk assessments and real-time monitoring of infrastructure, these technologies enhance decision-making processes and promote community safety. Moreover, the proactive design of disaster-resilient urban environments contributes to sustainable development and improved quality of life for residents. As urban areas continue to grow and face increasing risks from natural disasters, the application

of ML and DL will play a crucial role in creating safer and more resilient communities.

Challenges and future directions in hazards prediction using ML and DL

The application of ML and DL to hazard prediction presents several significant challenges that must be addressed to improve model performance and reliability. One of the most pressing issues is data scarcity and quality. High-quality, labeled data are often difficult to obtain, especially for rare events like earthquakes or tsunamis, which are critical for building accurate models. Furthermore, many datasets are imbalanced, and dominated by non-hazardous conditions, which can lead to biased predictions and poor generalization when predicting rare but high-impact events. To address this, techniques like data augmentation, synthetic data generation, and transfer learning are being explored to improve the quantity and diversity of training data (Tao et al., 2024). Another major challenge is the high computational demands associated with training DL models on large-scale datasets, such as those generated by satellite imagery or sensor networks. These models require substantial computational resources, which can limit their use in real-time hazard prediction scenarios. To mitigate this, researchers are developing model compression techniques such as pruning and quantization, as well as distributed computing frameworks that can scale model training across multiple nodes, reducing the time and resources needed for large-scale hazard prediction tasks (Mizrahi et al., 2024).

In addition to computational limitations, the issue of model interpretability remains a critical barrier to the broader adoption of ML and DL in hazard prediction. Complex deep learning models are often criticized for being "black boxes," where it is difficult to understand how they arrive at their predictions. This lack of transparency can be problematic in high-stakes situations like disaster response, where decision-makers need to trust and comprehend the reasoning behind model outputs (Liu et al., 2024a, 2024b). Enhancing model interpretability through techniques such as attention mechanisms, or using explainable AI tools like SHAP values, is an area of active research aimed at making these models more transparent and trustworthy. Generalizability across different types of hazards is also a significant concern. Models trained on one hazard, such as flooding, may not perform well when applied to other hazards like wildfires or earthquakes due to differences in the underlying data and hazard characteristics. Developing models that can generalize across multiple hazard types using multi-task learning or transfer learning techniques could help overcome this limitation, allowing for more flexible and adaptable predictive models that work across diverse hazard scenarios (Laino et al., 2024).

Lastly, ethical considerations surrounding bias in ML/DL models must be addressed, particularly the risk of bias in predictions that could disproportionately affect vulnerable populations. Since ML models are trained on historical data, they can inherit the biases present in that data, potentially leading to unequal outcomes in hazard response and resource allocation. Ensuring fairness, accountability, and transparency in the development and deployment of these models is essential, particularly in high-stakes fields like disaster management, where lives and property are at risk (Turner et al., 2024). Overall, while ML and DL have demonstrated considerable promise for improving hazard prediction, these challenges ranging from data quality and computational demands to model interpretability and ethical considerations must be addressed. Future research and development should focus on improving the scalability, transparency, and fairness of these models, as well as enhancing their ability to generalize across various hazard types, ultimately making them more robust tools for safeguarding lives and infrastructure in the face of disasters (Sköld Gustafsson et al., 2024).

Conclusion

In conclusion, the integration of ML and DL techniques into hazard assessment represents a significant advancement in predictive analytics. These technologies provide robust frameworks for processing and analyzing complex datasets, enabling more accurate and timely predictions of various hazards, including natural disasters, environmental risks, and public health threats. The ability to uncover patterns and relationships within vast amounts of data has transformed traditional hazard assessment methodologies, allowing for enhanced risk evaluation and management. Moreover, the diverse range of algorithms and architectures available ranging from decision trees and support vector machines to convolutional neural networks and recurrent neural networks affords researchers the flexibility to tailor models to specific hazards and datasets. This adaptability ensures that hazard assessments are not only precise but also contextually relevant, addressing unique regional and temporal dynamics. Despite the promise of ML and DL in hazard prediction, challenges remain. Issues such as data quality, model interpretability, and the need for continuous learning to adapt to evolving hazard patterns must be addressed to optimize their effectiveness. Furthermore, the ethical implications of automated decision-making in hazard management necessitate careful consideration to ensure that these technologies are deployed responsibly and equitably.

Ultimately, the continued exploration of machine learning and deep learning in hazard assessment has the potential to significantly enhance our preparedness and response strategies. As these technologies advance, they will play a critical role in building resilient communities capable of mitigating the impacts of hazards, thereby safeguarding lives and livelihoods. Future research should focus on overcoming existing challenges, exploring hybrid approaches, and fostering collaboration among stakeholders to maximize the benefits of ML and DL in hazard assessment.

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Declarations

Conflict of interest The authors declare no competing interests.

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